

# Curriculum Vitae

## Personal Data

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| Name      | Paul Schwahn                                                                                                     |
| Birthdate | 14. Feb. 1999                                                                                                    |
| Address   | IMECC, Universidade Estadual de Campinas<br>Rua Sérgio Buarque de Holanda 651<br>13083-859 Campinas-SP<br>Brazil |
| Email     | <code>pschwahn@uni-muenster.de</code>                                                                            |
| Website   | <a href="https://pschwahn.github.io">pschwahn.github.io</a>                                                      |
| ORCID     | <a href="https://orcid.org/0000-0003-0718-3949">0000-0003-0718-3949</a>                                          |
| Position  | Postdoctoral researcher at the University of Münster                                                             |

## Education

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| 2014–2017 | B.Sc. studies at the University of Stuttgart                                                                                                                                                                            |
| Oct 2017  | Graduation with a B.Sc. in Mathematics at the University of Stuttgart.<br>Bachelor's thesis: <i>Scattering theory for automorphic functions</i><br><br>Award of the <i>Robert Bosch GmbH</i> for excellent B.Sc. degree |
| 2017–2020 | M.Sc. studies at the JLU Gießen and the University of Stuttgart                                                                                                                                                         |
| Apr 2020  | Graduation with a M.Sc. in Mathematics at the University of Stuttgart.<br>Master's thesis: <i>Einstein deformations on homogeneous spaces</i>                                                                           |
| Jan 2021  | Award of the <a href="#">Freunde der Mathematik an der Universität Stuttgart</a> (Friends of mathematics at the University of Stuttgart) for best M.Sc. degree                                                          |
| Jul 2021  | Award of the <a href="#">Freunde der Universität Stuttgart</a> (Friends of the University of Stuttgart) for extraordinary scientific accomplishments                                                                    |
| 2020–2023 | Doctoral studies in Mathematics under the supervision of Prof. Uwe Semmelmann                                                                                                                                           |
| Jul 2023  | Graduation <i>summa cum laude</i> with a Dr.rer.nat. at the University of Stuttgart. Doctoral thesis: <a href="#">Stability of Einstein metrics on homogeneous spaces</a>                                               |

## Work experience & Funding

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| 2016–2020 | Student assistant/tutor for various mathematics courses at the University of Stuttgart                                                                                                                                                                                                                                |
| 2020–2023 | Research associate/teaching assistant at the Institute of Geometry and Topology, University of Stuttgart. Partially funded by <a href="#">SPP 2026 Geometry at Infinity</a><br><br>Assistance for the courses <i>Analysis</i> ; <i>Geometry for Teachers</i> ; <i>Advanced Mathematics</i> (for engineering students) |

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| 2023–2024 | Postdoctoral researcher at <i>Laboratoire de mathématiques d'Orsay</i> , Université Paris-Saclay. Funded by the <i>BRIDGES</i> collaboration, <a href="#">ANR-21-CE40-0017</a> . Partially funded by the <i>Procope program</i> , Project No. 48959TL |
| 2024–2026 | Postdoctoral researcher at IMECC, Unicamp. Fellow of <i>FAPESP</i> , <a href="#">Project 2024/08127-4</a> , under the <i>BRIDGES</i> collaboration                                                                                                    |
| 2026–2028 | Research associate at the Mathematical Institute, University of Münster. Funded by a DAAD <a href="#">PRIME fellowship</a>                                                                                                                            |
| 2026–2027 | Visiting researcher (12 months) at the Mathematical Institute, University of Oxford. Part of the DAAD <a href="#">PRIME fellowship</a>                                                                                                                |

### Schools, seminars, research stays

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| Sep 2016 | <a href="#">Ferienakademie</a> (summer school) of FAU Erlangen-Nürnberg, TU Munich and University of Stuttgart                   |
| Oct 2020 | Working seminar at Philipps-Universität Marburg, Department of Mathematics                                                       |
| Jul 2021 | Summer school course <a href="#">Riemannian foliations and Lie group actions</a> , Scuola Matematica Interuniversitaria, Cortona |
| May 2022 | Research stay at CUNY with Renato Bettiol                                                                                        |
| Sep 2022 | Block seminar <a href="#">Construction and degeneration of Einstein 4-manifolds</a> , University of Regensburg                   |
| Nov 2022 | Oberwolfach Graduate Seminar <a href="#">Geometry and Topology of Compact Homogeneous Spaces</a>                                 |
| Jun 2024 | Summer school of the <a href="#">BRIDGES meeting 2024</a> , Institut d'Études Scientifiques de Cargèse                           |
| Feb 2025 | <a href="#">MAM III</a> on Quaternion-Kähler manifolds and the LeBrun–Salamon Conjecture, Philipps-Universität Marburg           |

### Invited seminar talks

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| Sep 2021 | Differential Geometry Seminar, Universidad Nacional de Córdoba                                     |
| Nov 2021 | <a href="#">Geometry &amp; Topology Seminar</a> , University of Stuttgart                          |
| May 2022 | <a href="#">CUNY Geometric Analysis Seminar</a>                                                    |
| May 2022 | <a href="#">Geometry Seminar</a> , University of Florence                                          |
| Jul 2022 | <a href="#">London Geometry &amp; Topology Seminar</a>                                             |
| May 2023 | <a href="#">Riemannian Topology Seminar</a> , University of Stuttgart                              |
| May 2023 | Online talk at the <a href="#">Virtual Seminar on Geometry with Symmetries</a>                     |
| Jul 2023 | <a href="#">Working group seminar</a> of the differential geometry group, University of Regensburg |
| Oct 2023 | <i>Séminaire Géométrie Topologie Dynamique</i> , Université Paris-Saclay                           |

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| Dec 2023 | <a href="#">Research Seminar Differential Geometry and Analysis</a> of Philipps-Universität Marburg           |
| Mar 2024 | Geometry seminar of Université Claude-Bernard Lyon                                                            |
| Apr 2024 | <a href="#">ULB geometry seminar</a> , Université Libre de Bruxelles                                          |
| Jul 2024 | <a href="#">Geometry &amp; Topology Seminar</a> , University of Stuttgart                                     |
| Jul 2024 | <a href="#">Seminar Geometry</a> , KU Leuven                                                                  |
| Oct 2024 | <i>Seminário SP de Geometria</i> , São Carlos (joint meeting of ICMC, IME, UF-SCar, Unicamp)                  |
| Nov 2024 | Geometry seminar of IMECC, Unicamp                                                                            |
| Mar 2025 | <a href="#">Geometry seminar</a> of ICMC (São Carlos), University of São Paulo                                |
| Aug 2025 | Geometry seminar of IME (São Paulo), University of São Paulo                                                  |
| Nov 2025 | <a href="#">Seminário de Geometria Diferencial</a> , IMPA (Rio de Janeiro)                                    |
| Nov 2025 | Geometry seminar of Universidad Nacional de Córdoba                                                           |
| Nov 2025 | <a href="#">Seminario de Geometría UNS</a> , Universidad Nacional del Sur (Bahía Blanca)                      |
| Apr 2026 | <a href="#">Geometry &amp; Topology Seminar</a> , University of Stuttgart                                     |
| Aug 2026 | <a href="#">Colloquium</a> of the Department of Mathematical Physics, IF (São Paulo), University of São Paulo |
| Oct 2026 | <a href="#">Geometry &amp; Topology Seminar</a> , University of Stuttgart                                     |
| Oct 2026 | Online talk at the <a href="#">String Geometry Online Seminar</a>                                             |

### Conference posters

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| Jul 2021 | Poster and <a href="#">online talk</a> at <a href="#">Mathematical Congress of the Americas</a>                              |
| Mar 2024 | Poster at the conference <a href="#">Special Holonomy and Geometric Structures on Complex Manifolds</a> , IMPA               |
| Jul 2026 | Poster at the conference <a href="#">Differential Geometry 2026: A Tribute to Francesco Mercuri</a> , University of Cagliari |

### Conference talks

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| Aug 2022 | <a href="#">Spaces, Structures, Symmetries</a> , Università degli Studi di Bari Aldo Moro                                                 |
| Jun 2023 | BRIDGES workshop <a href="#">Special geometries and gauge theories</a> , University of Pau                                                |
| Jul 2023 | <a href="#">Einstein Spaces and Special Geometry</a> , Institut Mittag-Leffler (invited)                                                  |
| Jul 2024 | BRIDGES meeting in <a href="#">Gauge Theory, extremal structures and stability</a> , Institut d'Études Scientifiques de Cargèse (invited) |
| Jul 2024 | <a href="#">Junior Meeting Einstein Geometry and Special Holonomy</a> , CAU Kiel (invited)                                                |
| Jun 2025 | Problem-solving workshop <a href="#">Computational Geometric Analysis</a> , City University of New York (invited)                         |

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| Oct 2025 | <a href="#">CICM</a> (18th Conference on Intelligent Computer Mathematics), University of Brasilia                                    |
| Oct 2025 | <a href="#">Workshop on Symmetries in Geometry and Physics</a> , Universidad Nacional San Antonio de Abad del Cusco (invited)         |
| May 2026 | <a href="#">Campinas–Miami Mathematical Meeting</a> , IMECC, Unicamp (invited)                                                        |
| Jul 2026 | <a href="#">Workshop on Exceptional Structures and the Standard Model</a> , Higgs Centre for Theoretical Physics, Edinburgh (invited) |

### Organization of events

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| Sep 2023 | <a href="#">Sesquinar</a> , a student seminar about tensors, University of Stuttgart                                                    |
| Oct 2023 | International conference <a href="#">The Crazy World of Arthur L. Besse: a workshop on Einstein manifolds</a> , University of Stuttgart |
| Sep 2024 | <a href="#">Semi-LEVEN-ear</a> , a student seminar about formalization of mathematics, University of Stuttgart.                         |

### Miscellaneous

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| 2015–2020 | Commitment in the <a href="#">Faculty Student Group</a> in Stuttgart                                                                                                                                |
| 2022–2023 | Elected representative of doctoral students in the Faculty Council (Mathematics and Physics), Chairman of Faculty Student Council and member of the <a href="#">Student Parliament</a> in Stuttgart |

### Skills and interests

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| Research    | Riemannian geometry, curvature, special holonomy, geometric analysis, manifolds with symmetries, Lie theory |
| Languages   | English (fluent), French (very basic), German (native), Portuguese (basic)                                  |
| Programming | C++, LiE, Matlab, Python, SageMath                                                                          |
| Hobbies     | Climbing, Piano, Pen&Paper RPGs, Game development, Formalization of mathematics/computer-assisted proofs    |

Münster, September 30, 2026